

STARS & GALAXIES - QUICK REVISION GUIDE

Overview of star life cycles, galaxies, and other galactic features.

STARS

What Are Stars?

Huge spheres of hot gas (mostly hydrogen and helium).

- Produce energy through nuclear fusion
- Classified by size, temperature, and color
- Color indicates temperature (blue = hottest, red = coolest)

STAR LIFE CYCLE

1. Stellar Birth - Nebula

A large cloud of gas and dust where stars begin forming.

- Gravity pulls matter together
- Dense regions become protostars

2. Main Sequence Star

Longest stage of a star's life.

- Hydrogen fuses into helium
- Stable balance between gravity pulling inward and pressure pushing outward
- The Sun is a medium main-sequence star

3. After Main Sequence

Depends on star's mass:

A. Low / Medium Mass Stars (like the Sun)

Red Giant Stage

- Star expands as core contracts
- Outer layers cool → red color

Planetary Nebula

- Outer layers drift away into space

White Dwarf

- Hot, dense stellar core left behind
- Slowly cools over billions of years

Black Dwarf (theoretical)

- Final cooled state of a white dwarf

- None exist yet (universe not old enough)

B. High Mass Stars

Red Supergiant

- Massive expansion
- Heavy elements form in the core

Supernova

- Massive explosion that spreads elements into space
- Creates the heaviest elements (gold, uranium, etc.)

Then it becomes:

Neutron Star

- Extremely dense object left after the explosion
- Made almost entirely of neutrons

Black Hole

- If the core is massive enough
- Gravity so strong not even light escapes

GALAXIES

What Are Galaxies?

Huge collections of billions of stars, gas, dust, and dark matter.

- Held together by gravity
- Often contain star-forming regions, nebulae, and black holes
- Our Sun belongs to the Milky Way Galaxy

TYPES OF GALAXIES

1. Spiral Galaxies

Flat, rotating disks with spiral arms.

- Spiral arms rich in gas and dust → active star formation
- Central bulge with older stars
- The Milky Way is a barred spiral galaxy

Examples: Milky Way, Andromeda

2. Elliptical Galaxies

Round or oval-shaped; little structure.

- Mostly older stars
- Very little gas or dust
- Minimal new star formation
- Can range from small to incredibly massive

3. Irregular Galaxies

No defined shape.

- Often formed through galaxy collisions or gravitational distortion
- High rates of star formation
- Mix of old and new stars

4. Lenticular Galaxies (S0 type)

A blend of spiral and elliptical features.

- Disk like a spiral, but without well-defined arms
- Low gas and dust
- Mostly older stars

OTHER IMPORTANT GALACTIC FEATURES

Black Holes

- Many galaxies contain a supermassive black hole at the center
- Milky Way's central black hole: Sagittarius A

Dark Matter

- Invisible material that makes up most of a galaxy's mass
- Detected through gravitational effects

Star Clusters

- Open clusters: young stars, loose grouping
- Globular clusters: old stars, tightly packed

Nebulae

- Birthplaces of stars (e.g., Orion Nebula)
- Also remnants of dead stars (e.g., Crab Nebula)